Types of digital circuits

1. Combinational circuits
2. Sequential circuits

Combinational circuits:

- The output of the combinational circuits depend on one the present inputs
- The feed back path is not present in the combinational circuit
- In combinational circuits, memory elements are not required
- The clock signal is not required for combinational circuits
- The combinational circuits is simple to design
- Combinational circuit are
  - Half and full adder
  - Half and full Subtractor
  - Binary Adder Subtractor
  - Decimal to BCD converter

Sequential circuit:

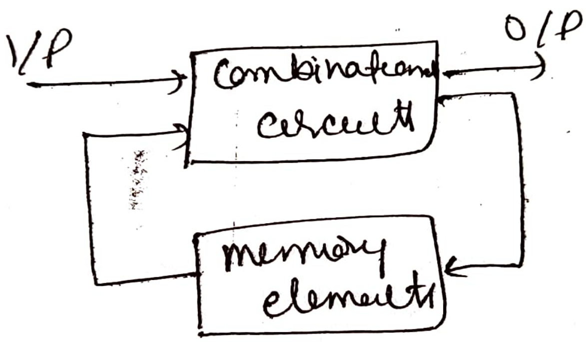
- The o/p of the sequential circuits depend on both present i/p and present state (previous o/p)
- The f/b path is present in the sequential circuits
- In the sequential circuit, memory elements play an important role and requre.
- The clock signal is required for sequential circuit
- It is not simple to design a sequential circuit

Sequential circuits are

- flip/flops
- Registers
- Counters
- Latches etc

→ Special types of circuits that has a series of inputs and O/P.

→ A sequential circuit doesn't need to always contain a Combinational circuit. So the sequential circuit can contain only the memory element



Types of Sequential Circuits

Asynchronous sequential circuits : Clock sig are not used by these circuits, operated throu pulses. to change state of circuits, The inter state is changed when the i/p variable is chan The unclocked f/f or time delayed memory ele are A.S. circuits → similar to Combinational circuit with f/b.

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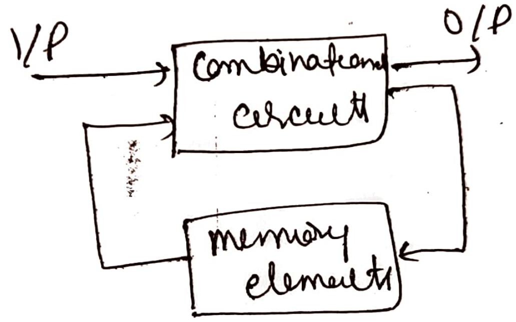

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f triggering

+ve edge triggering
-ve edge triggering

+ve edge triggering   -ve edge triggering

Basics of flip flops :

→ A circuit that has two stable states is treated as a f/f. These stables states are used to store binary data that ~~the~~ can be changed by applying varying inputs. The flip flop are the fundamental building blocks of digital systems. f/f and latches are examples of data storage elements.

Types of f/f

1. SR flip flop
2. J-K flip flop
3. D flip flop
4. T flip flop

S-R f/f

→ The S-R f/f is most common f/f used in the digital system. ~~The SR f/f when set input is high~~

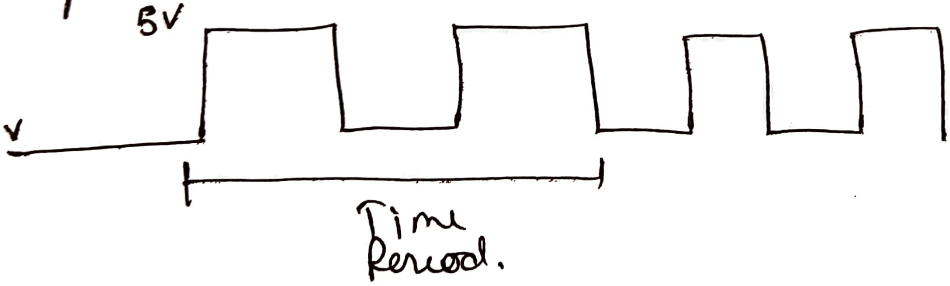
→

② Synchronous sequential circuits :

In synchronous sequential circuits, sepn memory element's State is done by the clock digital signal. The o/p is stored in either f/f or uses latches (m/s devices) The synchronization ica of o/p is done with either only negative edges of the clock signal or only positive edges.

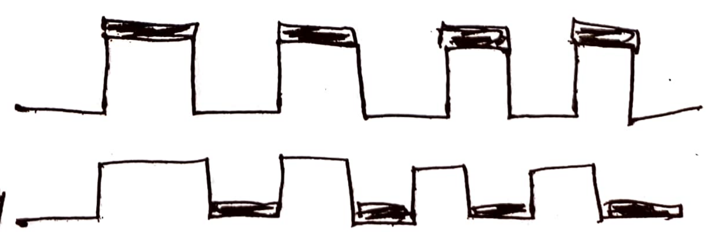
Clock signal and Triggering :

Clock signal : periodic signal in which ON time and OFF time need not be the same. When ON time and OFF time of the clock signal a square wave is used to represent the clock signal.

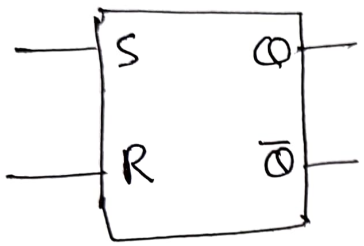


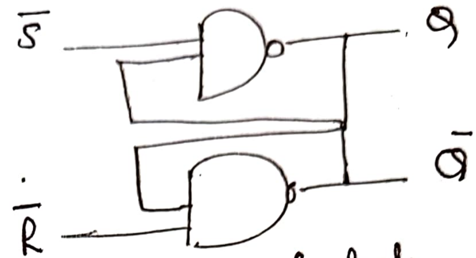
Types of Triggering :

Level triggering
+ve level triggering
-ve level triggering



R-S f/f : The RS f/f is used to temporarity hold or store information until it is needed. Single R.S flip flop will store one binary digit either a '1' or '0', Storing 4 digit binary number would require four R-S f/f





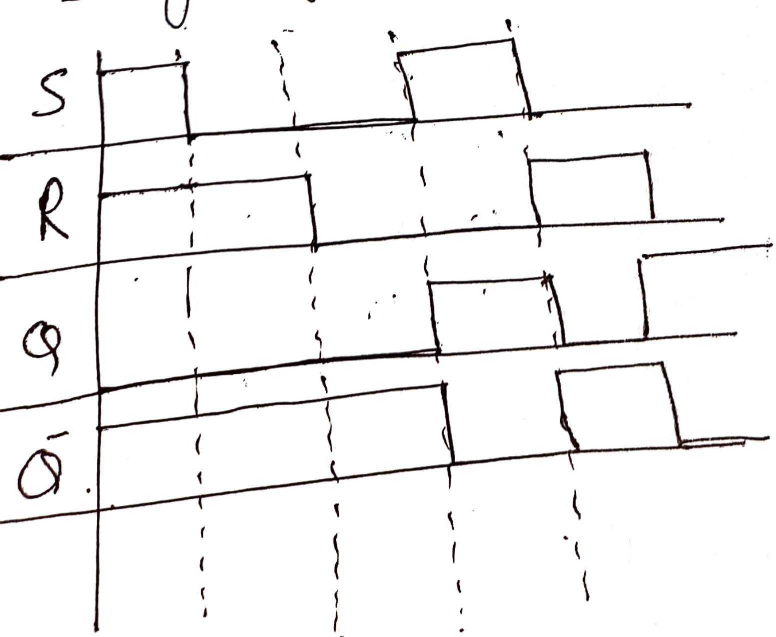
Catch

Characteristic table

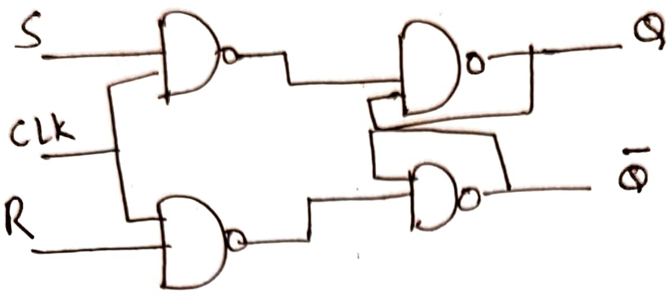
S	R	Q	$\bar{Q}$	O/P.
0	0	*	*	latched
0	1	0	1	reset
1	0	1	0	Set
1	1	X 0/1	X 1/0	Jammed.

* Previous state

Timing diagram



SR f/f : A flip flop, on other hand, is a synchronous circuit and also known as gated or clocked SR latch



o/p changed only when an active clock signal will be given

Truth table

S	R	Q	$\bar{Q}$
0	0	0	1
0	1	0	1
1	0	1	0
1	1	x	x

Characteristic table

Present i/p		Present State	Next state
S	R	Qt	Q(t+1)
0	0	0	0
0	1	0	
1	0		
1	1		

Characteristic table

Present i/p		Present State	Next State
S	R	Qt	Q(t+1)
0	0	0	0
0	1	0	1
0	0		
0	1		
1	0		
1	1		
1	0		
1	1		

S	R	Q	$\bar{Q}$
0	0	0	1
0	1	0	1
1	0	1	0
1	1	X	X

...racteristic table

S	R	Q_t	Q_{t+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	X
1	1	1	X

X – undefined states

S \ RQ	00	01	11	10
0		1		
1	1	1	X	X

— S

R'Q

$$\boxed{Q(t+1) = S + R'Q(t)}$$

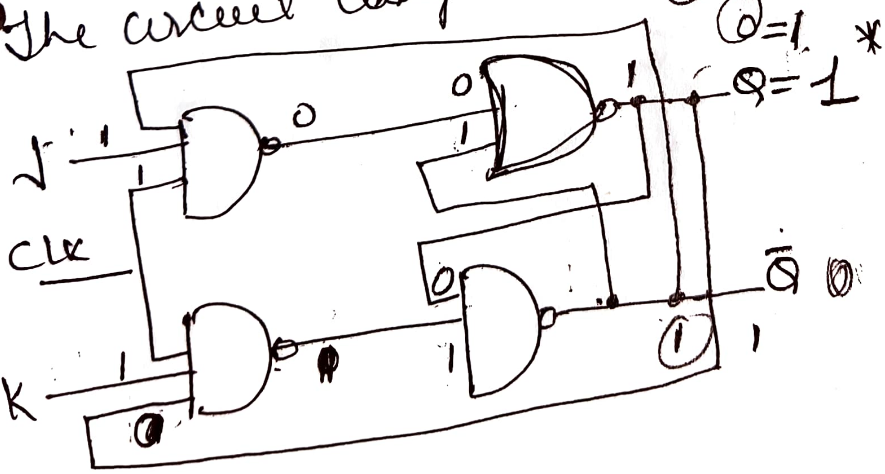
JK flip flop :

JK flip flop is modified version of SR f/f

JK flip flop is operates with only +ve clock or -ve clock

* used to remove undefined states

The circuit diagram



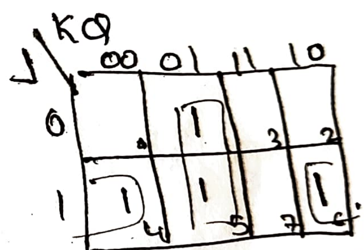
Truth table of J-K f/f

J	K	Q_{t+1}
0	0	Q_t
0	1	0
1	0	1
1	1	$\overline{Q_t}$

* Q_t & Q_{n+1}' are p... and next state
* Can be used for on... digital ...es ...at four fun:
  - Hold
  - Reset
  - Set
  - Complement of Present State

Characteristic table :

J	K	Q_t	Q_{t+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0



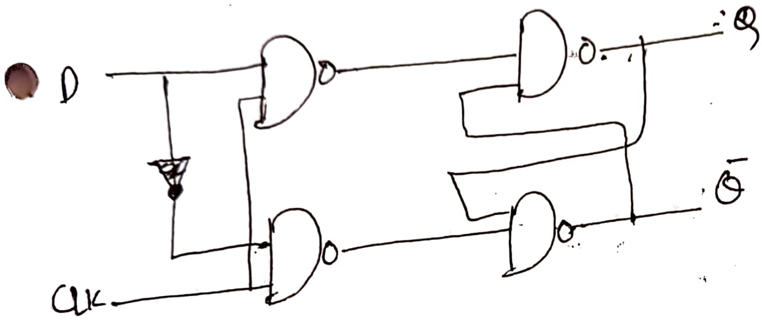
$\bar{K}Q$ $\quad J\bar{Q}$

$$\boxed{Q_{t+1} = \bar{K}Q + J\bar{Q}}$$

Name of letter   circuit. A digital

X ——— X ——— X

D f/f   widely, The D f/f is mostly used in Shift registers, Counters and input Synchronization



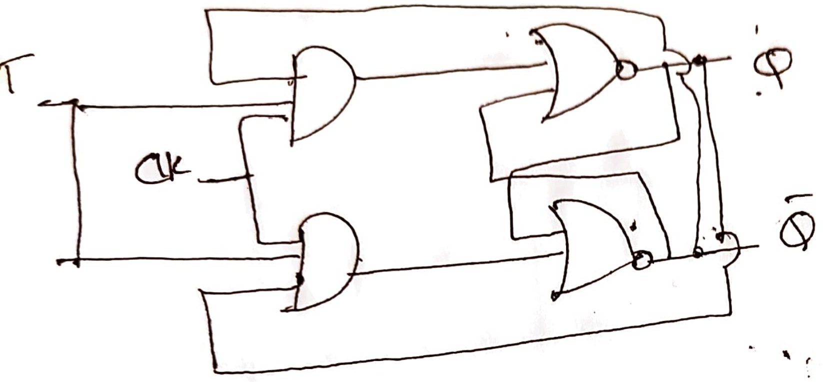
CK.	D	Q	Q'
0	0	0	1
1	0	0	1
0	1	0	1
1	1	1	0

2) T f/f

Toggle f/f   Just like JK f/f T f/f is used There are only single i/p with clock. The T f/f is constructed by connecting both of the i/p JK f/f together as a single input. The T f/f is Toggle f/f

Truth table

T	Y	Y(t+1)
0	0	0
1	0	1
0	1	1
1	1	0



0980 friday

1182

1330 → 12 Jan
1310 → 20 Jan
1328 → 12 Jan
1295 → 6 Jan
1314 → 20 Jan.

Counters : It is sequential circuit. A digital circuit which is used for a counting pulses is known counters. Counter is widest application of flip flops. It is group of flip flops with a clock signal applied. Counters are of two types ①

→ Asynchronous or ripple counters.
→ Synchronous Counters.

① **Asynchronous counters** :

These counter donot use universal clock, only first flip flop is driven by main clock and the clock input of ~~reset~~ of the following flip flop is driven by O/P of previous flip flops.

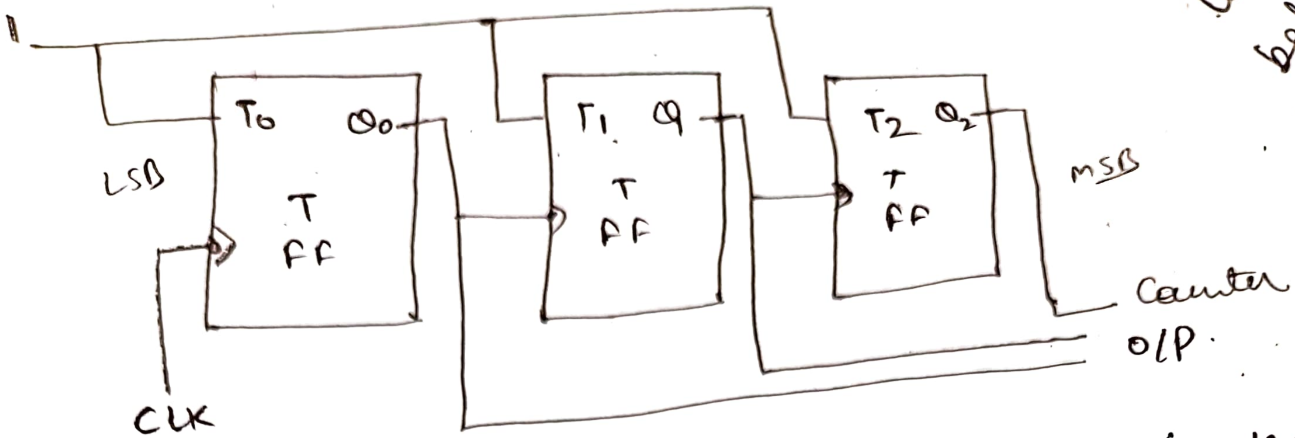
→ Asynchronous Binary up counters
→ " Binary down counters

• An 'N' bit binary counter of 'N' T flip flop. If the counter counts from 0 to 2^N-1, then it is called binary up counter.
→ Similary, if the counter counts down from 2^N-1 to 0, then it is called as binary down counter.

Asynchronous Binary up counter :

AN 'N' bit asynchronous binary up counter consist of 'N' T flip flop. It ~~consist~~ count from 0 to 2^N-1. The block diagram of 3 bit Async. binary up counter is shown.

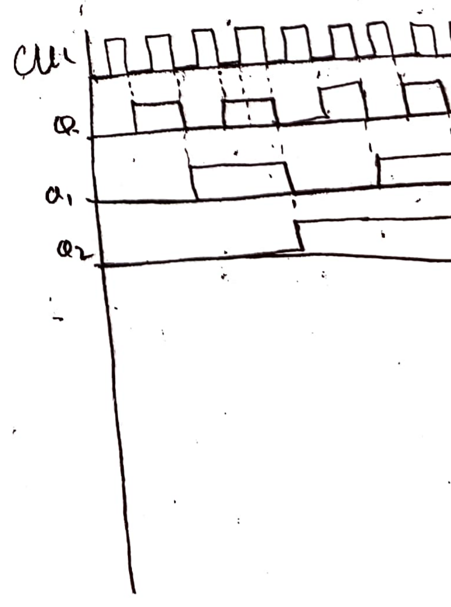
counts
below



→ The 3 bit asynchronous binary up counter contains three T flip flops and the T i/p of all the flip flops are connected to '1'. All these f/f are negative edge triggered but the o/p change asynchronously. The clock signal is directly applied to the first T f/f. So the output of first T f/f toggles for every negative edge of clock signal.

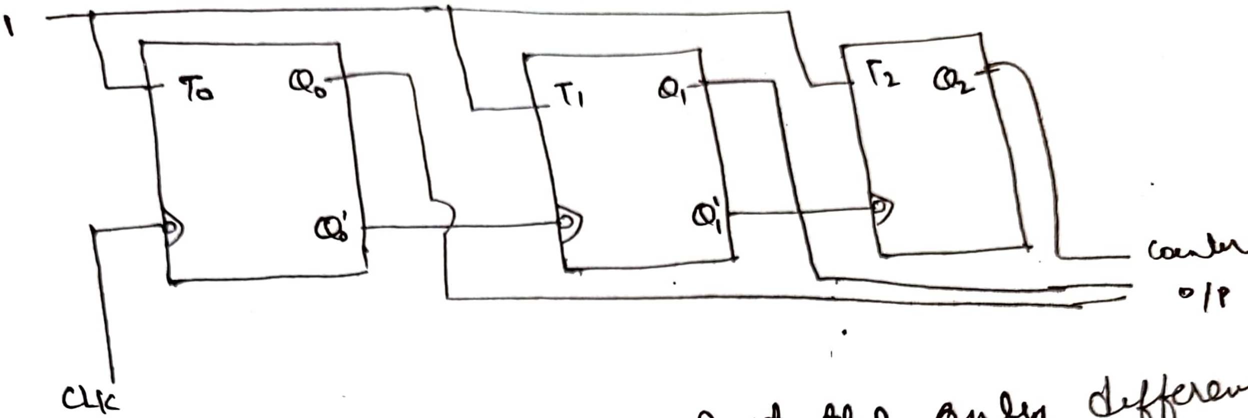
→ Assume the initial status of T f/f from rightmost to leftmost is $Q_2 Q_1 Q_0 = 000$. Here Q_2 & Q_0 are MSB & LSB respectively. We can understand working of 3 bit asynch binary counter.

No. of Neg. edge of clock	Q_0	Q_1	Q_2
0	0	0	0
1	1	0	0
2	0	1	0
3	1	1	0
4	0	0	1
5	1	0	1
6	0	1	1
7	1	1	1



ronous Binary Down Counter ②

It counts from $2^N - 1$ to 0. The block diagram shown is below



Same as the Binary up counter. But the only difference is the instead of connecting the normal o/p of one stage flip flop as clock signal to next stage flip flop, connect the complement o/p of one stage flip flop as clock signal for next stage flip flop.

H/f as clock signal from 1 to 0 is same as the

- Complemented o/p goes from 1 to 0 is same as the normal o/p goes from 0 to 1
- Initial status is $Q_2 Q_1 Q_0 = 000$

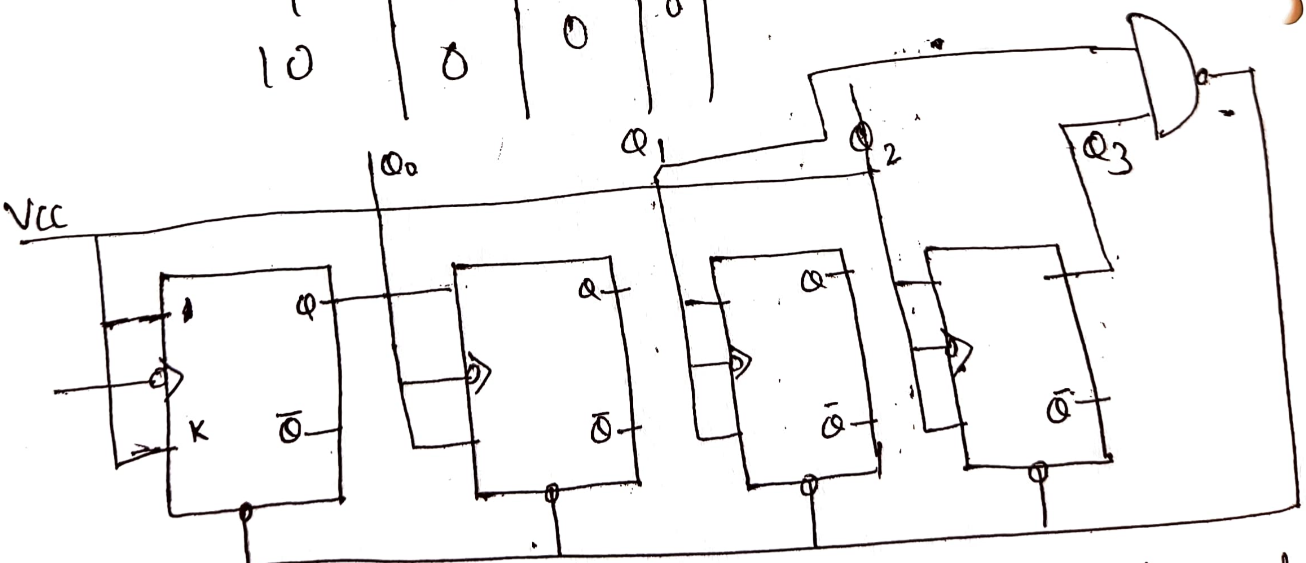
-ve CLK	Q_0	Q_1	Q_2
0	0	0	0
1	1	1	1
2	0	1	1
3	1	0	1
4	0	0	1
5	1	1	0
6	0	1	0
7	1	0	0

Q_2	Q_1	Q_0
0	0	0
1	1	1
1	1	0
1	0	1
1	0	0
0	1	1
0	1	0
0	0	1
0	0	0

Decade Counter : A decade counter co[illegible] different states and then reset to its init[illegible] states. A simple decade counter will count [illegible] 0 to 9 but we can also make decade counters which can go through any ten states b/w 0 to 15 for four bit counter)

Clock Pulse	Q_3	Q_2	Q_1	Q_0
	0	0	0	0
0	0	0	0	1
1	0	0	1	0
2	0	0	1	1
3	0	1	0	0
4	0	1	0	1
5	0	1	1	0
6	0	1	1	1
7	1	0	0	0
8	1	0	0	1
9	0	0	0	0
10				

If we see Q_3 & Q_1 are '1' in 1010 we give NAND of those two bits to clear input then counter will be clear at 10 and again start from begining.



We see from circuit diagrams that we have used nand for Q_3 and Q_1 and feeding this to clear input line because binary representation of 10 is 1010

[illegible] LSB bit applied first.

...chronous MOD Counters : A Counter can have ... possible counting states eg. Mod-16 for a 4-bit counter, (0-15) making it ideal for use in frequency Division application. But it is also possible to use the Basic asynchronous counter configuration to construct special counter with counting states less than their maximum O/P number. For example Modulo or MOD counters.

→ This achieved by forcing the counter to reset itself to zero at a pre-determined value producing a type of asynchronous ~~value~~ counter that has truncated sequences. Then an n-bit counter that counts up to its maximum modulus (2^n) is called a full sequence counter and n bit counter whose modulus is less than the maximum possible is called truncated counter.

1055 9-16 Medical

1093 Swap. 4/1/23

1094

The number of flip flop used in a ripple counter is depends upon the no. of states of counter (ex mod 4, MOD2) The no. of O/P states of counter is called Modulus or MOD of counter. The maximum number of states that are counter can have is 2^n where n represents the no. of flip flops used in counter

→ Can be easily designed by T f/f or D f/f

→ low speed circuits

→ They are used as Divide by n counters, which divide the input by n, where n is an integer

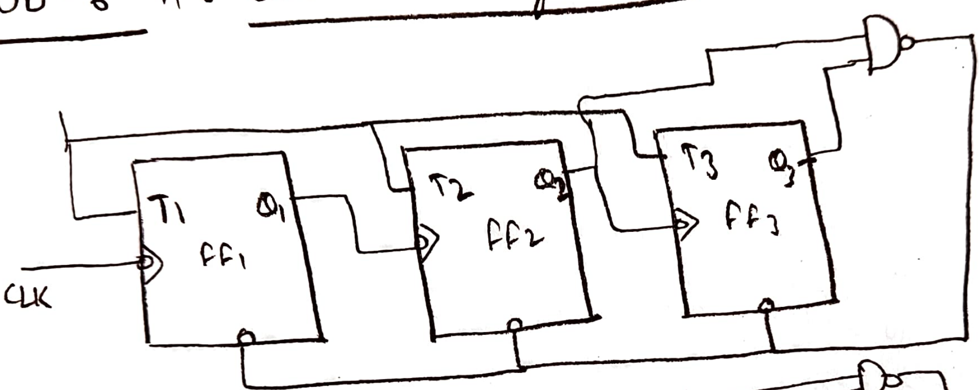
Disadvantages

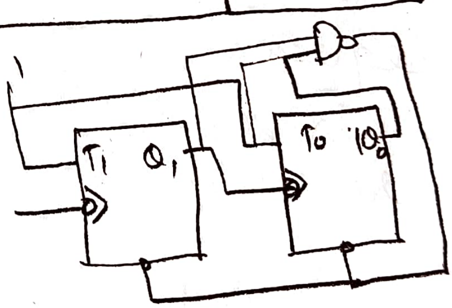
Require resynchronization
need extra circuitry
For high clock frequencies counting error may occur due to propagation delay

Application

frequency dividers
low power application
Low noise emission
designing asynchronous decade counter
Also used in Ring counter & Johnson counter

MOD-6 A.S. Counter using TFFs





After Pulse	Q_3	Q_2	Q_1
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0

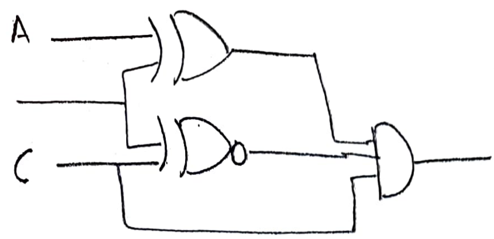
NAND

4MOD

	Q_1	Q_0
0	0	0
1	0	1
2	1	0
3	1	1

32910 6789

9. input condition required for $X = 1$



D

$\begin{pmatrix} A & B & C \\ 0 & 1 & 1 \end{pmatrix}$

A	B	B
1	1	1

Q Determine the O/P frequency for a frequency division circuit that contain 12 f/f with in put clock freq 20.48 MHz

$$\frac{20.48 \times 10^6}{2^{12}} = \frac{20.48 \times 10^6}{4096} = 5000 \text{ Hz}$$

...based on Flip Flops & Register. ... counter can have ... 4-bit

...o is J.K. f/f mode to toggle

Q 0010 is serially entered (right most bit first) into a 4 bit SIPO that is initially clear. O/P at Two clock pulse

C	Q_3	Q_2	Q_1	Q_0
0	0	0	1	0
1	0	0	0	1
2	1	0	0	0

0010

420
942
995
1000

Q Assume that 4 bit serial in / serial out shift register is initially clear. We wish to store the nibble 1100. What will be the 4 bit pattern after second clock pulse

[1 | 1 | 0 | 0]

	Q_3	Q_2	Q_1	Q_0
0	0	0	0	0
1	0	0	0	0
2	0	0	0	0
3	1	0	0	0
4	1	1	0	0

Q Serial in / Parallel out [1111]

Q_3	Q_2	Q_1	Q_0
1	1	1	0
1	1	1	1
1	1	1	1
1	1	1	1
0	1	1	1

0111

Q How many f/f required to make 32 MOD counter.
5

$2^n - 1$
0 to 31
4 ff → 0 to 16
5 → 0 to $2^n - 1$
0 to 2^5
$2 \times 2 \times 2 \times 2 \times 2$ = 32

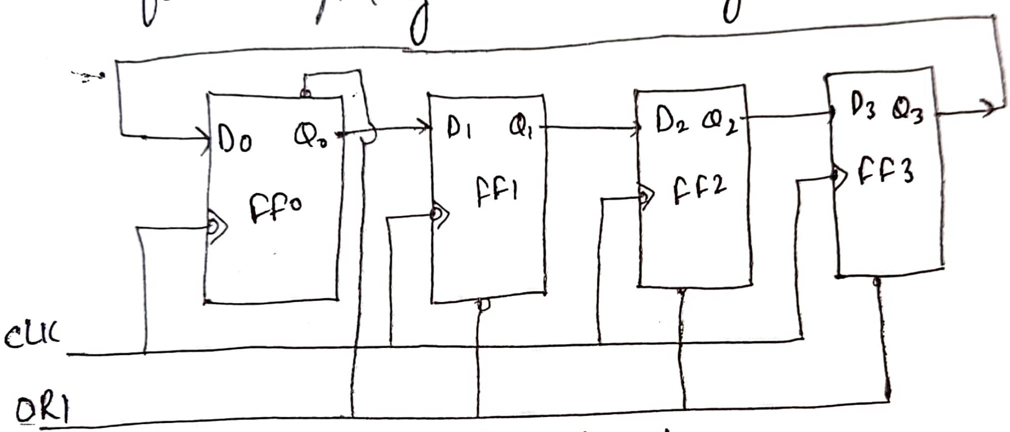
Q Difference b/w register & a counter - register has no specific sequence of state

1055	4160	4,01,1301

counter : A ring counter is a typical application of shift register. The ring counter is almost the same as the shift counter. The only change is that O/P of last flip flop is connected to the I/P of the first flip flop in the case of the ring counter but in the case of shift register it is taken as output. Except for this, all the other things are the same.

No. of States in Ring Counter = No. of flip flop used

So for designing a 4 bit Ring Counter we need 4 f/f



RING Counter.

In this diagram, we can see that the clock pulse (CLK) is applied to all the flip flop simultaneously. Therefore, it is synchronous counter. Also, here we use overriding input (ORI) for each flipflip. Preset (PR) and Clear (CLR) are used as ORI. When PR is 0, then the O/P is 1. And when CLR is 0, then O/P is 0. Both PR and CLR are active low signal that always works in value 0.

PR = 0, Q = 1

CLR = 0 Q = 0

These two values are always fixed. They are independent of the value of input D and the clock pulse (Clk). Working Here, ORI is connected to PReset (PR) in FF-0, and it is connected to clear (CLR) in FF-1, FF-2 and FF-3. Thus, O/P Q = 1 is generated at FF-0, and rest of the flip generates O/P Q = 0. This O/P Q = 1 at FF = 0 is known as Preset 1, which is used to form the ring in Ring Counter.

ORI	CLK	Q_0	Q_1	Q_2	Q_3
0	X	1	0	0	0
1	0	0	1	0	0
1	0	0	0	1	0
1	0	0	0	0	1
1	0	1	0	0	0

This Presented 1 is generated by making ORI Low and that time clk becomes dont care. After that ORI is made high and apply low clock pulse signal as the clock (CLK) is negative edge triggerd. After that at each clock pulse, the presented 1, is shifted to the next flip flop and thus forms a Ring. From the above table, we can say that there are 4 states in a 4- Ring Counter.

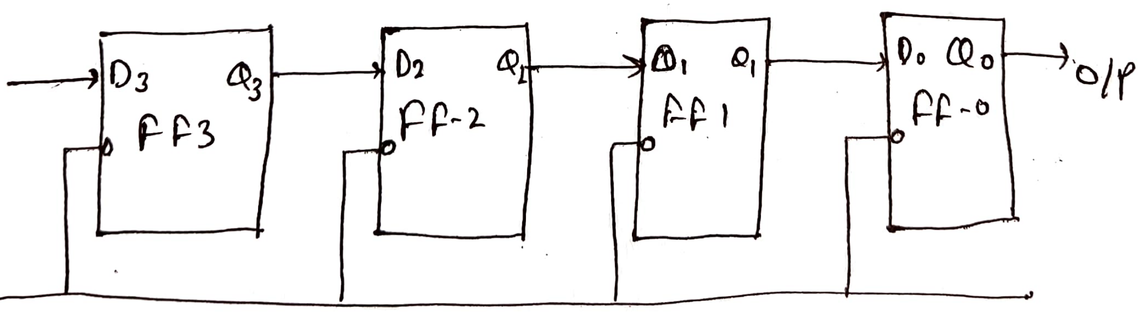
(2) (1)

Registers : f/f is a 1 bit memory cell which can be used for storing the digital data. To increase the storage capacity in terms of number of bits we have to use a group of flip flop. Such a group of flip-flop is known as Register. The n bit Register will consist of n number of flip flop and it is capable of storing an n-bit word.

The binary data in a register can be moved within the register from one flip flop to another. The register that allow such data transfer are called as Shift register. There are four mode of operation of a Shift register.

→ Serial input serial output
→ Serial input parallel output
→ parallel input serial output
→ parallel input parallel output

1) SISO :



Let all flip flop be initially in the reset condition

$Q_3 = Q_2 = Q_1 = Q_0 = 0$.

Let suppose a four bit binary number 1111 is made into register, this number should be applied to Din bit with LSB bit applied first.

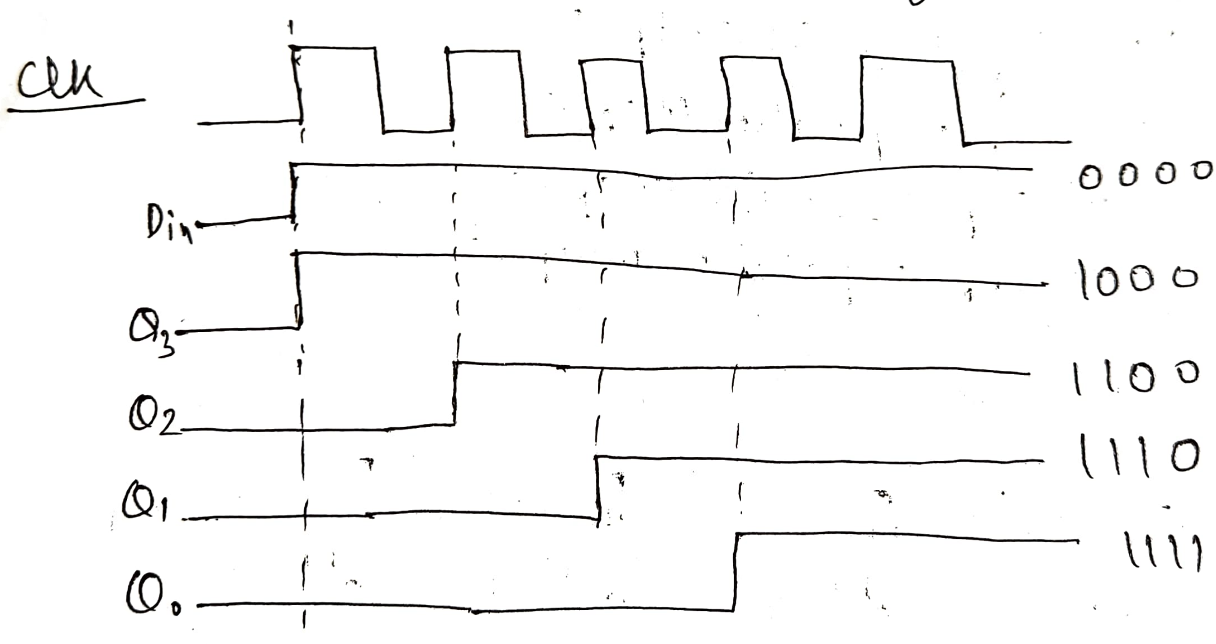
D_3 is connected to serial Data D_{in}. O/P of is Q_3 connected to next f/f D_2.

Intial $Q_3 Q_2 Q_1 Q_0 = 0000$

Truth Table :

Clk	D_{in}	Q_3	Q_2	Q_1	Q_0
0	0	0	0	0	0
1	1	1	0	0	0
1	1	1	1	0	0
1	1	1	1	1	0
1	1	1	1	1	1

Direction of data



SISO → $2N-1$ → $2(8)-1 = 7$

SIPO → N → 4

PISO → $N-1$

PIPO → 1

4

1303

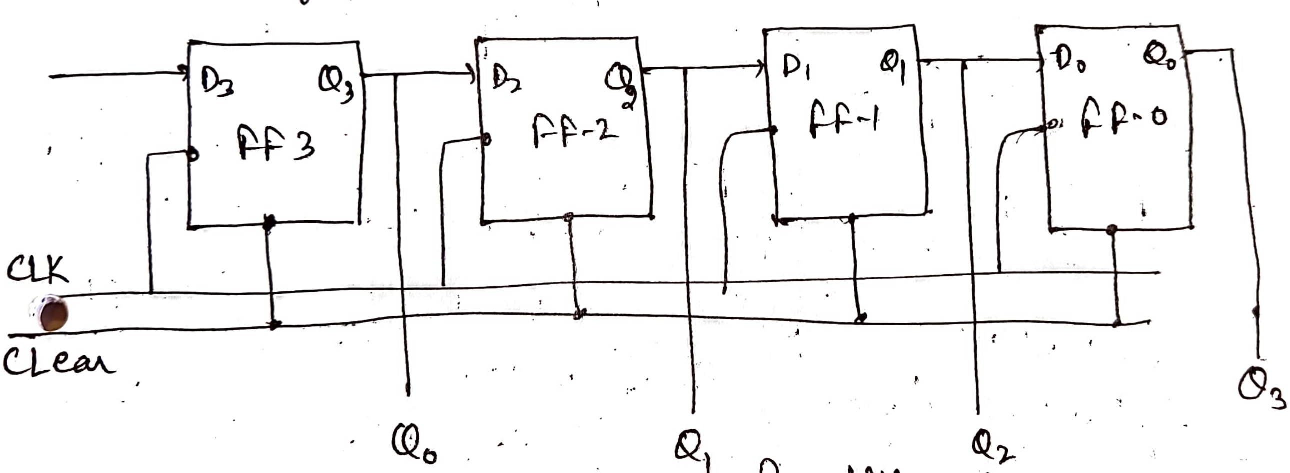
bit memory cell which ... data. To increase ...

SIPO: Data is entered serially and taken out in parallel fashion. Data is loaded bit by bit, The O/P is disabled as long as the data is loading.

→ As soon as the data loading gets completed, all flip flops contain their required data, the O/P are enabled so that all the loaded data is made available over all the O/P line at the same time

→ 4 clock cycles are required to load of four bit word. Hence the speed of operation of SIPO mode is same as that of SISO mode.

Block Diagram

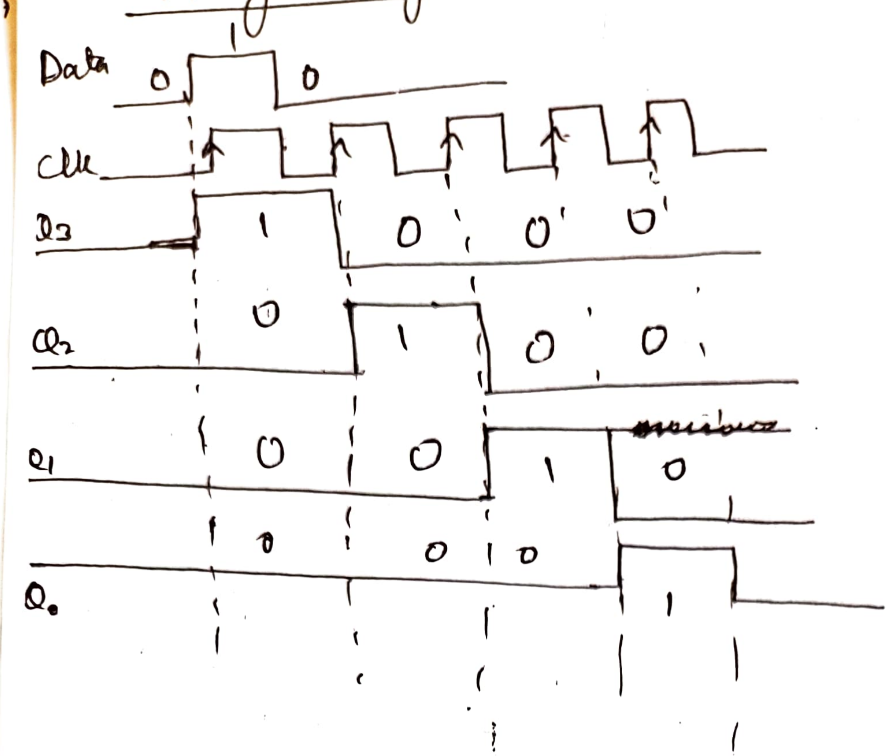


Truth table

CLK	Q_3	Q_2	Q_1	Q_0	
0	0	0	0	0	0
1	1	0	0	0	1
2	0	1	0	0	
3	0	0	1	0	
4	0	0	0	1	
5	0	0	0	0	

(2)

Timing Diagram :



3) Parallel input serial o/p (PISO)

→ Data bits are entered in parallel fashion

→ four bit PISO

→ o/p of previous flipflop is connected to the input of the next one via a combinational circuit

→ The binary input word B_0 B_1 B_2 B_3 is applied through a same combinational circuit.

→ There two modes in which this circuit can work namely - shift mode or load mode.

→ Load mode : when the Shift/Load bar is low (0) and the AND gate 2, 4 and 6 become active they will pass B_1, B_2, B_3 bits to the corresponding flip flop. On the high going edge of clock, the binary input B_0, B_1, B_2, B_3 will get loaded into the corresponding f/f. Thus parallel loading take place.